Analytic Geometry for Computer Scientists

Lecture Notes

Prepared as mathematical lecture notes

May 16, 2026

Abstract

These notes present analytic geometry as a mathematical language for computer science. The emphasis is on precise definitions, derivations, formulas, and computational interpretation. The central objects are points, vectors, lines, planes, hyperplanes, distances, projections, transformations, and geometric tests. The notes are suitable for a mathematics lecture for students of computer science, especially before or alongside linear algebra, computer graphics, robotics, computer vision, optimization, numerical methods, and computational geometry.

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How to read these notes

Analytic geometry translates geometric questions into algebraic calculations. Instead of asking only whether a point is visually above a line, we ask for the sign of a determinant. Instead of drawing the shortest distance from a point to a plane, we compute an orthogonal projection. Instead of rotating an object by intuition, we multiply its coordinate vector by a matrix.

For computer scientists this is not only a theoretical convenience. Most computational systems that manipulate geometry use analytic descriptions internally. Pixels, meshes, cameras, animations, collision tests, optimization procedures, and neural network embeddings all use coordinates, vectors, matrices, dot products, projections, and distances.

Key idea

Analytic geometry is the bridge between visual geometry and algorithms. A geometric statement becomes useful in computer science when it can be represented, computed, tested, and made numerically stable.

The notes are organized in a progressive way:

1. Points and coordinates.
2. Vectors and vector operations.
3. Lines, planes, and hyperplanes.
4. Distances, angles, and projections.
5. Affine transformations and coordinate frames.
6. Barycentric coordinates and geometric primitives.
7. Conics, quadrics, and implicit geometry.
8. Computational geometry tests and numerical issues.

The level is intentionally mathematical. Algorithms appear when they clarify why a formula matters computationally.

1 Coordinates, points, and vectors

1.1 Coordinates as numerical representation of geometry

A point in the plane can be represented by an ordered pair

$$P = (x, y) \in \mathbb{R}^2.$$

A point in three-dimensional space is represented by

$$P = (x, y, z) \in \mathbb{R}^3.$$

More generally, a point in n-dimensional Euclidean space is represented by

$$P = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n.$$

The important idea is that a point is geometric, while its coordinates depend on the chosen coordinate system. The same physical or geometric point may have different coordinates in different frames.

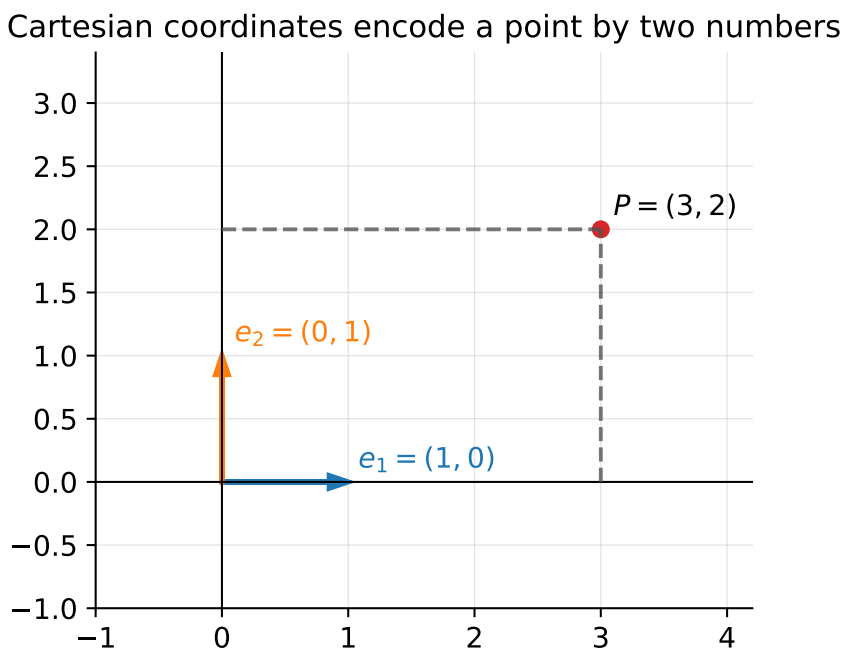

Figure 1: A point is encoded by coordinates relative to chosen axes.

Definition 1.1 (Point)**.** A point is an element of an affine space. In elementary analytic geometry over $\mathbb{R}$, we identify points with tuples of real numbers after choosing a coordinate system.

Definition 1.2 (Vector)**.** A vector represents a displacement. In $\mathbb{R}^n$, a vector is an ordered n-tuple

$$v = (v_1, v_2, \ldots, v_n).$$

It may be drawn as an arrow, but algebraically it is an element of the vector space $\mathbb{R}^n$.

A vector can be added to a point:

$$P + v = (p_1 + v_1, \ldots, p_n + v_n).$$

The difference of two points is a vector:

$$Q - P = (q_1 - p_1, \ldots, q_n - p_n).$$

This distinction is very important.

Common pitfall

A point is a location. A vector is a displacement. In code both may be stored as arrays of numbers, but the mathematical meaning is different. Adding two points has no intrinsic affine meaning, while subtracting two points gives a vector.

1.2 Vector operations

For vectors $u, v \in \mathbb{R}^n$ and scalar $\lambda \in \mathbb{R}$, define

$$u + v = (u_1 + v_1, \ldots, u_n + v_n),$$

$$\lambda u = (\lambda u_1, \ldots, \lambda u_n).$$

The zero vector is

$$0 = (0, 0, \ldots, 0).$$

The standard basis of $\mathbb{R}^n$ is

$$e_1 = (1, 0, \ldots, 0), \quad e_2 = (0, 1, 0, \ldots, 0), \quad \ldots, \quad e_n = (0, \ldots, 0, 1).$$

Every vector $v \in \mathbb{R}^n$ has the representation

$$v = v_1 e_1 + v_2 e_2 + \cdots + v_n e_n.$$

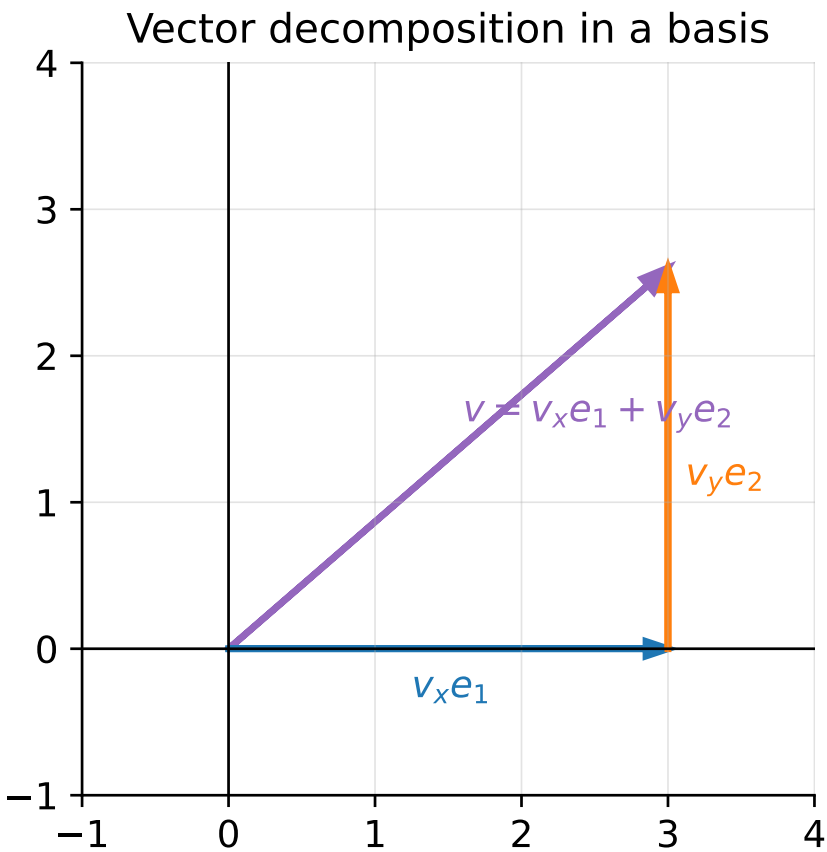

Figure 2: A vector decomposes into components in a chosen basis.

1.3 Linear combinations and spans

Definition 1.3 (Linear combination)**.** A vector w is a linear combination of vectors $v_1, \ldots, v_k$ if there exist scalars $\alpha_1, \ldots, \alpha_k$ such that

$$w = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k.$$

Definition 1.4 (Span)**.** The span of vectors $v_1, \ldots, v_k$ is the set of all their linear combinations:

$$\operatorname{span}(v_1, \ldots, v_k) = \{\alpha_1 v_1 + \cdots + \alpha_k v_k : \alpha_i \in \mathbb{R}\}.$$

In geometry, a span gives a line, plane, or higher-dimensional linear subspace through the origin. For example:

- If $v \neq 0$, then $\operatorname{span}(v)$ is a line through the origin.

- If $u, v \in \mathbb{R}^3$ are not parallel, then $\operatorname{span}(u, v)$ is a plane through the origin.

Computer science viewpoint

Spans appear in data science as feature subspaces, in graphics as local coordinate frames, and in numerical linear algebra as column spaces of matrices.

1.4 Affine combinations

Linear combinations are centered at the origin. Geometry often needs combinations of points that do not depend on the origin.

Definition 1.5 (Affine combination)**.** An affine combination of points $P_1, \ldots, P_k$ is an expression

$$\alpha_1 P_1 + \cdots + \alpha_k P_k$$

where

$$\alpha_1 + \cdots + \alpha_k = 1.$$

The condition that the coefficients sum to one makes the expression independent of the origin. For two points P, Q, the point

$$(1-t)P + tQ$$

lies on the line through P and Q. If $0 \le t \le 1$, it lies on the segment joining them.

2 Norms, distances, and the dot product

2.1 Euclidean norm

Definition 2.1 (Euclidean norm)**.** For $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$, the Euclidean norm is

$$\|v\| = \sqrt{v_1^2 + \cdots + v_n^2}.$$

The Euclidean norm measures length. It satisfies:

$$\|v\| \ge 0, \qquad \|v\| = 0 \iff v = 0,$$

$$\|\lambda v\| = |\lambda| \, \|v\| \,,$$

$$\|u + v\| \le \|u\| + \|v\| \,.$$

The last inequality is the triangle inequality.

2.2 Distance between points

The distance between points $P, Q \in \mathbb{R}^n$ is the norm of their difference:

$$\operatorname{dist}(P, Q) = \|Q - P\| \,.$$

In the plane:

$$\operatorname{dist}(P, Q) = \sqrt{(q_x - p_x)^2 + (q_y - p_y)^2}.$$

In three-dimensional space:

$$\operatorname{dist}(P, Q) = \sqrt{(q_x - p_x)^2 + (q_y - p_y)^2 + (q_z - p_z)^2}.$$

2.3 Dot product

Definition 2.2 (Dot product)**.** For $u, v \in \mathbb{R}^n$, the dot product is

$$u \cdot v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

The dot product is bilinear:

$$(au + bv) \cdot w = a(u \cdot w) + b(v \cdot w),$$

and symmetric:

$$u \cdot v = v \cdot u.$$

It also satisfies

$$v \cdot v = \|v\|^2 .$$

2.4 Angle formula

For nonzero vectors u, v, the angle $\theta \in [0, \pi]$ between them is defined by

$$u \cdot v = \|u\| \, \|v\| \cos\theta.$$

Therefore

$$\cos\theta = \frac{u \cdot v}{\|u\| \, \|v\|}.$$

Proposition 2.3 (Orthogonality criterion)**.** *For nonzero vectors u, v,*

$$u \perp v \iff u \cdot v = 0.$$

Proof. The vectors are perpendicular exactly when $\theta = \pi/2$. Since $\cos(\pi/2) = 0$, the angle formula gives $u \cdot v = 0$. Conversely, if $u \cdot v = 0$, then $\cos\theta = 0$, hence $\theta = \pi/2$. □

2.5 Projection onto a vector

Let $a \neq 0$. The projection of b onto a is the component of b in the direction of a. We seek a vector of the form ta such that the error

$$b - ta$$

is perpendicular to a. Thus

$$a \cdot (b - ta) = 0.$$

Expanding:

$$a \cdot b - t(a \cdot a) = 0.$$

So

$$t = \frac{a \cdot b}{a \cdot a}.$$

Therefore

$$\operatorname{proj}_a b = \frac{a \cdot b}{a \cdot a} a.$$

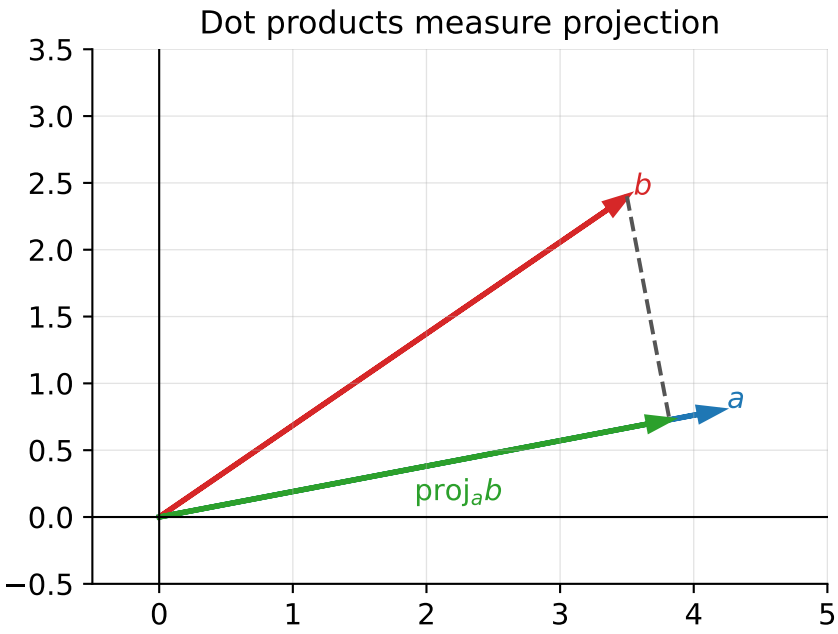

Figure 3: The projection of b onto a is found using the dot product.

Formula

For $a \neq 0$,

$$\operatorname{proj}_a b = \frac{a \cdot b}{\|a\|^2} a.$$

The perpendicular residual is

$$b - \operatorname{proj}_a b.$$

Computer science viewpoint

Projection is used in least squares, graphics lighting models, vector embeddings, dimensionality reduction, and collision detection. Many algorithms reduce a problem by decomposing a vector into parallel and perpendicular components.

3 Determinants, orientation, and area

3.1 The determinant in the plane

For vectors $u = (u_x, u_y)$ and $v = (v_x, v_y)$ in $\mathbb{R}^2$, define

$$\det(u, v) = \begin{vmatrix} u_x & v_x \\ u_y & v_y \end{vmatrix} = u_x v_y - u_y v_x.$$

Geometrically, $|\det(u, v)|$ is the area of the parallelogram spanned by u and v. The sign gives orientation:

- $\det(u, v) > 0$: turning from u to v is counterclockwise.
- $\det(u, v) < 0$: turning from u to v is clockwise.
- $\det(u, v) = 0$: the vectors are linearly dependent, so they are parallel or one is zero.

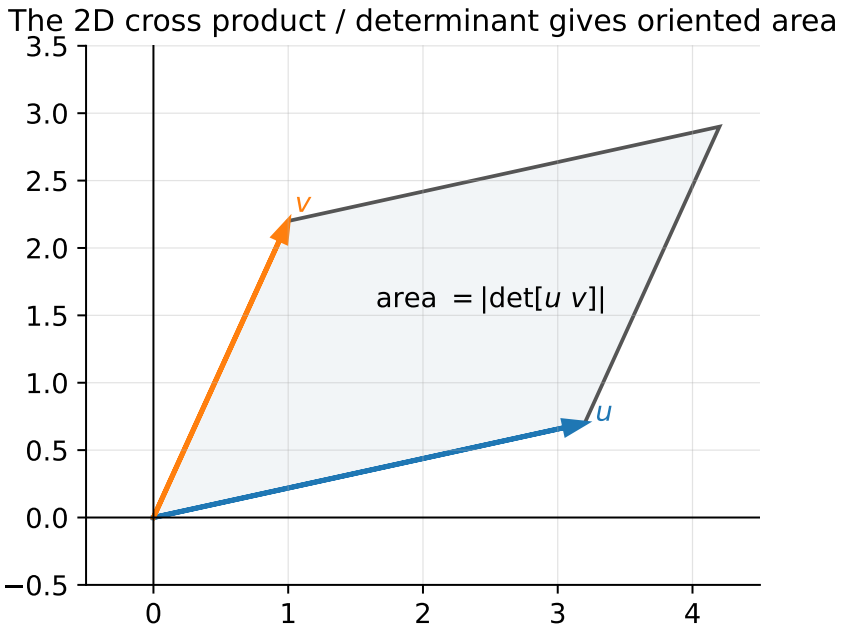

Figure 4: The determinant gives signed area in the plane.

3.2 Derivation of the area formula

The parallelogram generated by u and v is the image of the unit square under the linear map with matrix

$$A = \begin{pmatrix} u_x & v_x \\ u_y & v_y \end{pmatrix}.$$

The determinant is the signed area scaling factor of this linear map. Hence the area of the image of the unit square is $|\det A|$, which equals $|\det(u, v)|$.

3.3 Orientation test

Given three points $A, B, C \in \mathbb{R}^2$, define

$$\text{orient}(A, B, C) = \det(B - A, C - A).$$

Then:

$$\text{orient}(A, B, C) > 0$$

means that the ordered triple A, B, C makes a counterclockwise turn,

$$\text{orient}(A, B, C) < 0$$

means clockwise, and

$$\text{orient}(A, B, C) = 0$$

means collinear.

Listing 1: Orientation test for three planar points.

```
function orientation(A, B, C):
    return det(B - A, C - A)

if orientation(A, B, C) > 0:
    return "counterclockwise"
if orientation(A, B, C) < 0:
```

```
    return "clockwise"
return "collinear"
```

Computer science viewpoint

The orientation test is one of the basic primitives of computational geometry. It appears in convex hull algorithms, line segment intersection, polygon triangulation, rasterization, and mesh processing.

3.4 Signed area of a polygon

Let a polygon have vertices

$$p_0, p_1, \ldots, p_{m-1}$$

in cyclic order, where $p_i = (x_i, y_i)$. Define $p_m = p_0$. Its signed area is

$$A_{\text{signed}} = \frac{1}{2} \sum_{i=0}^{m-1} \det(p_i, p_{i+1}).$$

Equivalently,

$$A_{\text{signed}} = \frac{1}{2} \sum_{i=0}^{m-1} (x_i y_{i+1} - y_i x_{i+1}).$$

This is often called the shoelace formula.

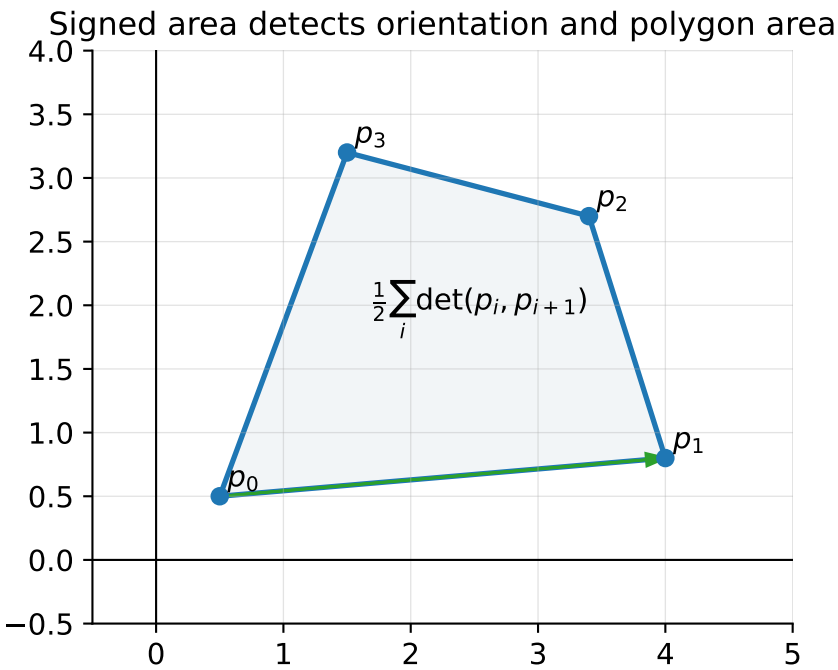

Figure 5: The signed area formula is a sum of determinant terms.

3.5 Cross product in three dimensions

For $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ in $\mathbb{R}^3$, the cross product is

$$u \times v = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix}.$$

It is perpendicular to both u and v:

$$(u \times v) \cdot u = 0, \qquad (u \times v) \cdot v = 0.$$

Its length is

$$\|u \times v\| = \|u\| \, \|v\| \sin \theta,$$

where θ is the angle between u and v. Thus $\|u \times v\|$ is the area of the parallelogram spanned by u and v.

Formula

In $\mathbb{R}^3$, the area of the triangle with vertices A, B, C is

$$\frac{1}{2} \|(B - A) \times (C - A)\| .$$

4 Lines in the plane and in space

4.1 Parametric form of a line

Let $P \in \mathbb{R}^n$ and let $d \in \mathbb{R}^n$ be a nonzero direction vector. The line through P in direction d is

$$L = \{P + td : t \in \mathbb{R}\}.$$

This is called the parametric equation of a line.

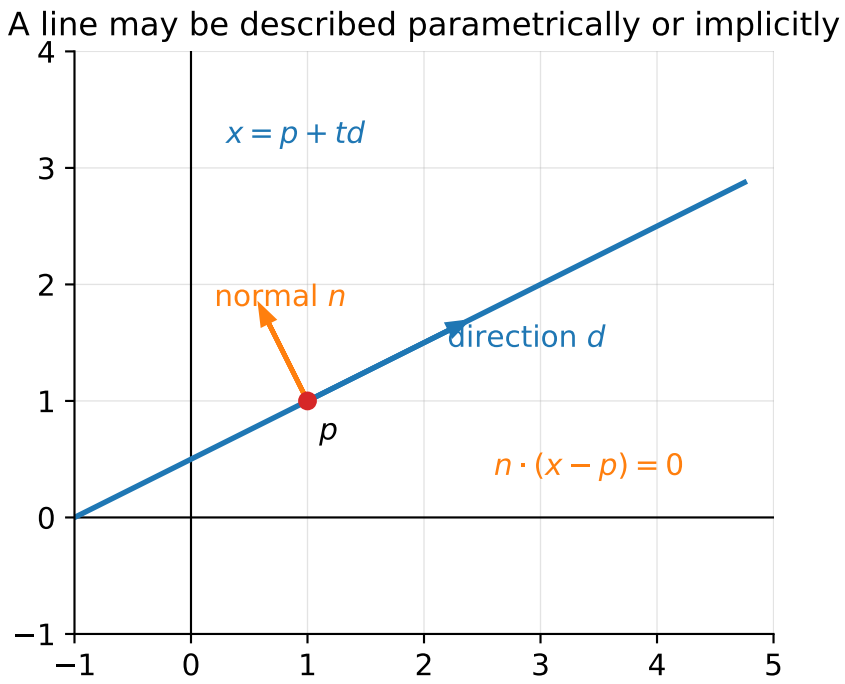

Figure 6: A line can be represented by a point and a direction, or by a normal equation.

4.2 Segments and rays

The segment from P to Q is

$$[P, Q] = \{(1 - t)P + tQ : 0 \leq t \leq 1\}.$$

Equivalently,

$$[P, Q] = \{P + t(Q - P) : 0 \leq t \leq 1\}.$$

The ray starting at P in direction $d \neq 0$ is

$$\{P + td : t \geq 0\}.$$

4.3 Implicit form of a line in the plane

A nonvertical or vertical line in $\mathbb{R}^2$ can be written uniformly as

$$ax + by + c = 0,$$

where $(a, b) \neq (0, 0)$. The vector

$$n = (a, b)$$

is normal to the line. If $X = (x, y)$, then the equation becomes

$$n \cdot X + c = 0.$$

If P is a point on the line, then $c = -n \cdot P$, so

$$n \cdot (X - P) = 0.$$

4.4 From parametric form to implicit form in $\mathbb{R}^2$

Suppose

$$L = \{P + td : t \in \mathbb{R}\},$$

where $P = (p_x, p_y)$ and $d = (d_x, d_y) \neq 0$. A normal vector to d is

$$n = (-d_y, d_x).$$

Thus an implicit equation is

$$n \cdot (X - P) = 0.$$

Explicitly,

$$-d_y(x - p_x) + d_x(y - p_y) = 0.$$

4.5 From implicit form to parametric form

Given

$$ax + by + c = 0,$$

a direction vector is

$$d = (-b, a),$$

because

$$(a, b) \cdot (-b, a) = 0.$$

To get a point P on the line, solve one simple equation. For example, if $b \neq 0$, set $x = 0$, giving

$$y = -\frac{c}{b}.$$

Then

$$P = \left(0, -\frac{c}{b}\right).$$

4.6 Line through two points

If $P \neq Q$, the line through P and Q is

$$L = \{P + t(Q - P) : t \in \mathbb{R}\}.$$

In $\mathbb{R}^2$, an implicit equation is

$$\det(Q - P, X - P) = 0.$$

This determinant is zero exactly when $Q - P$ and $X - P$ are linearly dependent, i.e. when X lies on the same line.

5 Planes and hyperplanes

5.1 Planes in $\mathbb{R}^3$

A plane in $\mathbb{R}^3$ can be represented parametrically by a point P and two nonparallel direction vectors u, v:

$$\Pi = \{P + su + tv : s, t \in \mathbb{R}\}.$$

If u and v are nonparallel, then

$$n = u \times v$$

is a normal vector to the plane. Hence the plane can also be written implicitly as

$$n \cdot (X - P) = 0.$$

If $n = (a, b, c)$, this becomes

$$a(x - p_x) + b(y - p_y) + c(z - p_z) = 0.$$

Equivalently,

$$ax + by + cz + d = 0.$$

5.2 Hyperplanes in $\mathbb{R}^n$

Definition 5.1 (Hyperplane)**.** A hyperplane in $\mathbb{R}^n$ is a set of the form

$$H = \{x \in \mathbb{R}^n : n \cdot x + c = 0\},$$

where $n \neq 0$.

The vector n is normal to the hyperplane. In $\mathbb{R}^2$, a hyperplane is a line. In $\mathbb{R}^3$, a hyperplane is a plane.

Computer science viewpoint

In machine learning, a linear classifier often uses a hyperplane

$$w \cdot x + b = 0.$$

The sign of $w \cdot x + b$ decides on which side of the decision boundary the point lies.

5.3 Half-spaces

A hyperplane separates $\mathbb{R}^n$ into two half-spaces:

$$H^+ = \{x : n \cdot x + c \geq 0\},$$

$$H^- = \{x : n \cdot x + c \leq 0\}.$$

Half-spaces are fundamental in optimization and computational geometry. A convex polygon in $\mathbb{R}^2$ can be represented as an intersection of half-planes. A convex polyhedron in $\mathbb{R}^3$ can be represented as an intersection of half-spaces.

6 Distances and projections

6.1 Point-to-line distance in $\mathbb{R}^2$

Let a line be given by

$$L = \{P + td : t \in \mathbb{R}\},$$

where $d \neq 0$. For a point Q, the closest point on the line has the form

$$P + t^* d.$$

We want the residual

$$Q - (P + t^* d)$$

to be perpendicular to d. Therefore

$$d \cdot (Q - P - t^* d) = 0.$$

Solving for t^*:

$$d \cdot (Q - P) - t^*(d \cdot d) = 0,$$

$$t^* = \frac{d \cdot (Q - P)}{d \cdot d}.$$

The closest point is

$$C = P + \frac{d \cdot (Q - P)}{d \cdot d} d.$$

The distance is

$$\operatorname{dist}(Q, L) = \|Q - C\|.$$

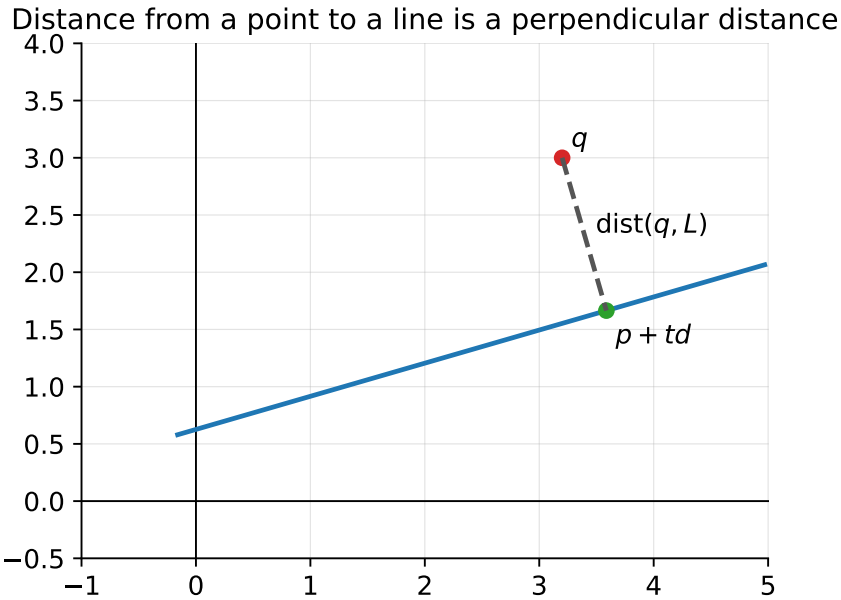

Figure 7: The shortest distance from a point to a line is perpendicular to the line.

6.2 Point-to-line distance using an implicit equation

Let

$$L = \{X \in \mathbb{R}^2 : ax + by + c = 0\}.$$

Let $n = (a, b)$. For a point $Q = (q_x, q_y)$, the signed distance to the line is

$$\frac{n \cdot Q + c}{\|n\|}.$$

The unsigned distance is

$$\operatorname{dist}(Q, L) = \frac{|aq_x + bq_y + c|}{\sqrt{a^2 + b^2}}.$$

Proof. The line consists of all points X satisfying $n \cdot X + c = 0$. Choose any point $P \in L$. Then $c = -n \cdot P$, and

$$n \cdot Q + c = n \cdot (Q - P).$$

The scalar projection of $Q - P$ onto the unit normal $\hat{n} = n/\|n\|$ is

$$(Q - P) \cdot \hat{n} = \frac{n \cdot (Q - P)}{\|n\|} = \frac{n \cdot Q + c}{\|n\|}.$$

Its absolute value is the perpendicular distance. □

6.3 Point-to-plane distance in $\mathbb{R}^3$

Let a plane be given by

$$\Pi = \{X : n \cdot X + d = 0\},$$

where $n \neq 0$. For $Q \in \mathbb{R}^3$,

$$\operatorname{dist}(Q, \Pi) = \frac{|n \cdot Q + d|}{\|n\|}.$$

The signed distance is

$$\frac{n \cdot Q + d}{\|n\|}.$$

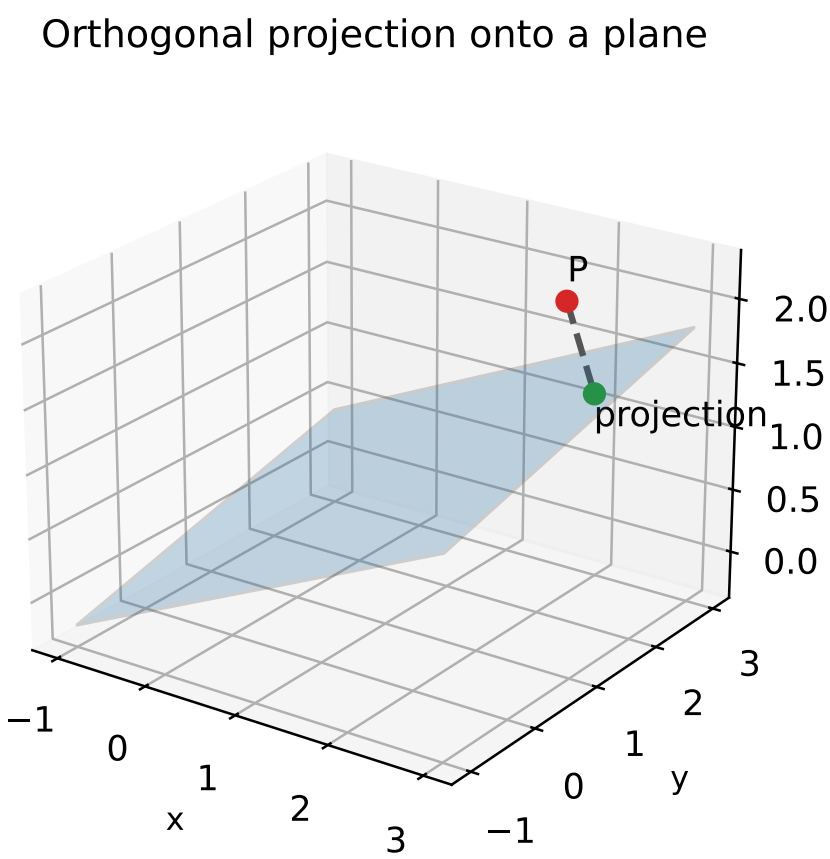

Figure 8: Orthogonal projection of a point onto a plane.

6.4 Projection onto a hyperplane

Let

$$H = \{x : n \cdot x + c = 0\}$$

with $n \neq 0$. Given $q \in \mathbb{R}^n$, the orthogonal projection of q onto H is

$$p = q - \frac{n \cdot q + c}{\|n\|^2} n.$$

Proof. The projected point lies on the line through q in direction n:

$$p = q - tn.$$

It must satisfy the hyperplane equation:

$$n \cdot (q - tn) + c = 0.$$

Thus

$$n \cdot q - t\,\|n\|^2 + c = 0,$$

so

$$t = \frac{n \cdot q + c}{\|n\|^2}.$$

Substituting gives the formula. □

6.5 Distance from a point to a segment

Let the segment be $[A, B]$, with $A \neq B$. Let $d = B - A$. The closest point on the infinite line is

$$A + t^* d, \qquad t^* = \frac{d \cdot (Q - A)}{d \cdot d}.$$

For the segment, the parameter is clamped to $[0, 1]$:

$$t = \min(1, \max(0, t^*)).$$

Then the closest point on the segment is

$$C = A + td,$$

and the distance is

$$\operatorname{dist}(Q, [A, B]) = \|Q - C\|.$$

Listing 2: Point-to-segment distance.

```
function distance_point_segment(Q, A, B):
    d = B - A
    t = dot(d, Q - A) / dot(d, d)
    t = clamp(t, 0, 1)
    C = A + t*d
    return norm(Q - C)
```

Computer science viewpoint

This formula is used in collision detection, nearest point queries, vector graphics, path planning, and distance field construction.

7 Intersections and incidence

7.1 Intersection of two lines in the plane

Let

$$L_1 : P + td, \qquad L_2 : Q + su,$$

where $d, u \neq 0$. We need

$$P + td = Q + su.$$

Rearrange:

$$td - su = Q - P.$$

This is a 2×2 linear system in unknowns t, s:

$$\begin{pmatrix} d_x & -u_x \\ d_y & -u_y \end{pmatrix} \begin{pmatrix} t \\ s \end{pmatrix} = Q - P.$$

If the determinant is nonzero, the lines meet in exactly one point. If the determinant is zero, the lines are parallel or coincident.

7.2 Intersection using determinants

In $\mathbb{R}^2$, the parameter t may be written as

$$t = \frac{\det(Q - P, u)}{\det(d, u)}$$

when $\det(d, u) \neq 0$. To derive this, take the determinant of both sides

$$td - su = Q - P$$

with u:

$$\det(td, u) - \det(su, u) = \det(Q - P, u).$$

Since $\det(u, u) = 0$, this gives

$$t \det(d, u) = \det(Q - P, u).$$

7.3 Segment intersection

Two segments $[A, B]$ and $[C, D]$ intersect if the corresponding line parameters satisfy

$$0 \leq t \leq 1, \qquad 0 \leq s \leq 1.$$

For robust computational geometry, orientation tests are often used instead.

For non-collinear cases, segments $[A, B]$ and $[C, D]$ intersect if and only if

$$\operatorname{orient}(A, B, C) \operatorname{orient}(A, B, D) < 0$$

and

$$\operatorname{orient}(C, D, A) \operatorname{orient}(C, D, B) < 0.$$

Collinear boundary cases require additional interval checks.

7.4 Line-plane intersection

Let a line be

$$X(t) = P + td$$

and a plane be

$$n \cdot X + c = 0.$$

Substitute the line into the plane equation:

$$n \cdot (P + td) + c = 0.$$

Thus

$$n \cdot P + t(n \cdot d) + c = 0.$$

If $n \cdot d \neq 0$, then

$$t = -\frac{n \cdot P + c}{n \cdot d}.$$

If $n \cdot d = 0$, then the line is parallel to the plane. It either does not intersect the plane or lies entirely in it.

Formula

Line-plane intersection:

$$X = P + td, \qquad n \cdot X + c = 0,$$

$$t = -\frac{n \cdot P + c}{n \cdot d} \quad \text{if } n \cdot d \neq 0.$$

8 Affine geometry

8.1 Why affine geometry matters

Euclidean geometry measures distances and angles. Affine geometry preserves lines, parallelism, and ratios along a line, but not necessarily lengths or angles. Computer graphics, image transformations, and geometric modeling rely heavily on affine maps.

Definition 8.1 (Affine map)**.** A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is affine if it has the form

$$T(x) = Ax + b,$$

where A is an $m \times n$ matrix and $b \in \mathbb{R}^m$.

If $b = 0$, the map is linear. If $b \neq 0$, the map includes a translation.

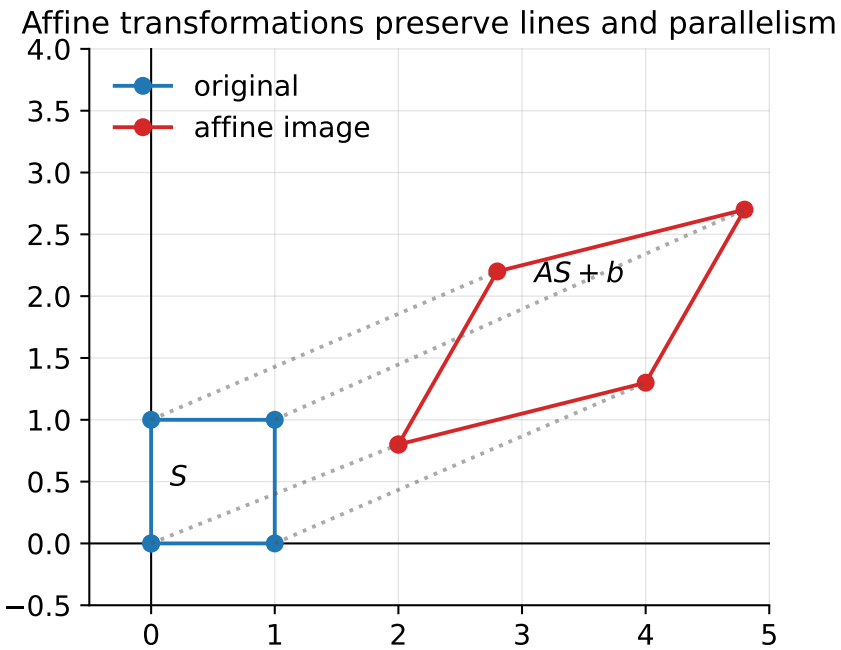

Figure 9: An affine transformation maps a square to a parallelogram and preserves parallelism.

8.2 Affine maps preserve affine combinations

Let $T(x) = Ax + b$, and let $\alpha_1 + \cdots + \alpha_k = 1$. Then

$$T(\alpha_1 P_1 + \cdots + \alpha_k P_k) = A(\alpha_1 P_1 + \cdots + \alpha_k P_k) + b.$$

By linearity of A:

$$= \alpha_1 AP_1 + \cdots + \alpha_k AP_k + b.$$

Since $\sum_i \alpha_i = 1$,

$$b = (\alpha_1 + \cdots + \alpha_k)b.$$

Therefore

$$T(\alpha_1 P_1 + \cdots + \alpha_k P_k) = \alpha_1(AP_1 + b) + \cdots + \alpha_k(AP_k + b),$$

so

$$T(\alpha_1 P_1 + \cdots + \alpha_k P_k) = \alpha_1 T(P_1) + \cdots + \alpha_k T(P_k).$$

Theorem 8.2. *Affine maps preserve affine combinations. Consequently, they map lines to lines, segments to segments, and planes to planes or lower-dimensional affine sets.*

8.3 Basic transformations in $\mathbb{R}^2$

Rotation by angle θ around the origin:

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Scaling:

$$S = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix}.$$

Shear in the x-direction:

$$H_x = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

Translation by $b = (b_x, b_y)$:

$$T(x) = x + b.$$

8.4 Homogeneous coordinates

Translations are affine but not linear in ordinary coordinates. Computer graphics represents affine maps as matrix multiplications using homogeneous coordinates.

A point $(x, y) \in \mathbb{R}^2$ is represented by

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

The affine map

$$T(x) = Ax + b, \qquad A = \begin{pmatrix} a & b_1 \\ c & d \end{pmatrix}, \qquad b = \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

is represented by

$$\begin{pmatrix} a & b_1 & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix}.$$

Then

$$\begin{pmatrix} a & b_1 & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} ax + b_1 y + t_x \\ cx + dy + t_y \\ 1 \end{pmatrix}.$$

Computer science viewpoint

Homogeneous coordinates allow a graphics pipeline to combine rotation, scaling, shear, and translation into one matrix product. This makes transformation composition fast and uniform.

8.5 Composition of transformations

If

$$T_1(x) = A_1 x + b_1, \qquad T_2(x) = A_2 x + b_2,$$

then

$$(T_2 \circ T_1)(x) = T_2(A_1 x + b_1) = A_2(A_1 x + b_1) + b_2.$$

So

$$(T_2 \circ T_1)(x) = (A_2 A_1)x + (A_2 b_1 + b_2).$$

Order matters: in general,

$$T_2 \circ T_1 \neq T_1 \circ T_2.$$

9 Coordinate frames and change of basis

9.1 Frames

A coordinate frame in $\mathbb{R}^n$ consists of an origin point O and a basis $(b_1, \ldots, b_n)$. A point P is represented by coordinates $\xi = (\xi_1, \ldots, \xi_n)$ if

$$P = O + \xi_1 b_1 + \cdots + \xi_n b_n.$$

Let

$$B = \begin{pmatrix} | & | & & | \\ b_1 & b_2 & \cdots & b_n \\ | & | & & | \end{pmatrix}.$$

Then

$$P = O + B\xi.$$

9.2 World coordinates and local coordinates

If P is known in local coordinates ξ, its world coordinates are

$$x = O + B\xi.$$

If B is invertible, local coordinates are obtained from world coordinates by

$$\xi = B^{-1}(x - O).$$

Formula

Frame conversion:

$$\boxed{x = O + B\xi}$$

from local to world coordinates, and

$$\boxed{\xi = B^{-1}(x - O)}$$

from world to local coordinates.

9.3 Orthonormal frames

A basis $(b_1, \ldots, b_n)$ is orthonormal if

$$b_i \cdot b_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

If B has orthonormal columns, then

$$B^T B = I,$$

so

$$B^{-1} = B^T.$$

This makes coordinate conversion efficient and numerically stable:

$$\xi = B^T(x - O).$$

Computer science viewpoint

Camera coordinate systems in graphics and vision are usually represented by orthonormal frames. Transforming a world point into camera coordinates is a change of frame.

10 Barycentric coordinates

10.1 Barycentric coordinates on a segment

For points A, B, any point on the line through them has the affine representation

$$P = (1-t)A + tB.$$

The numbers $(1-t, t)$ are barycentric coordinates with respect to A, B. The point is on the segment if

$$0 \le t \le 1.$$

10.2 Barycentric coordinates in a triangle

Let $A, B, C \in \mathbb{R}^2$ be non-collinear. A point P has barycentric coordinates (α, β, γ) with respect to triangle ABC if

$$P = \alpha A + \beta B + \gamma C, \qquad \alpha + \beta + \gamma = 1.$$

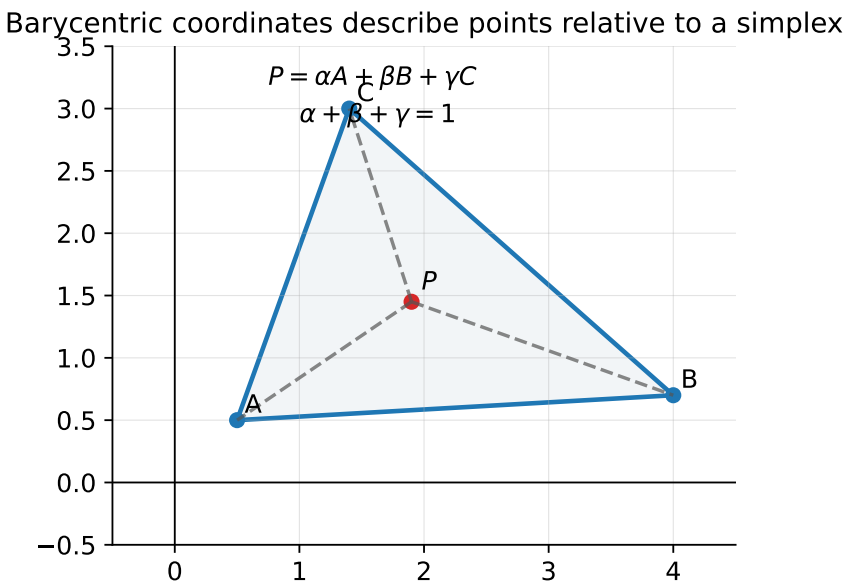

Figure 10: Barycentric coordinates express a point as an affine combination of triangle vertices.

10.3 Computing barycentric coordinates

Write

$$P = A + \beta(B - A) + \gamma(C - A).$$

Then

$$P - A = \beta(B - A) + \gamma(C - A).$$

This is a 2×2 linear system for β and γ. After solving, set

$$\alpha = 1 - \beta - \gamma.$$

Using determinants, define

$$D = \det(B - A, C - A).$$

Assume $D \neq 0$. Then

$$\beta = \frac{\det(P - A, C - A)}{D}, \qquad \gamma = \frac{\det(B - A, P - A)}{D},$$

and

$$\alpha = 1 - \beta - \gamma.$$

10.4 Point-in-triangle test

A point P lies inside or on the boundary of triangle ABC if and only if its barycentric coordinates satisfy

$$\alpha \geq 0, \qquad \beta \geq 0, \qquad \gamma \geq 0.$$

It lies strictly inside if all three inequalities are strict.

Computer science viewpoint

Barycentric coordinates are used in rasterization, texture mapping, interpolation over triangular meshes, finite element methods, and geometric modeling.

10.5 Interpolation on triangles

Suppose a scalar value is known at triangle vertices:

$$f(A) = f_A, \qquad f(B) = f_B, \qquad f(C) = f_C.$$

For a point

$$P = \alpha A + \beta B + \gamma C,$$

the linearly interpolated value is

$$f(P) = \alpha f_A + \beta f_B + \gamma f_C.$$

The same formula works for colors, texture coordinates, normals, and other vertex attributes.

11 Circles, conics, and quadrics

11.1 Circle

A circle with center $C = (a, b)$ and radius $r > 0$ is

$$(x - a)^2 + (y - b)^2 = r^2.$$

In vector form:

$$\|X - C\|^2 = r^2.$$

A point is inside the circle if

$$\|X - C\|^2 < r^2,$$

on the circle if equality holds, and outside if

$$\|X - C\|^2 > r^2.$$

11.2 Sphere

A sphere in $\mathbb{R}^3$ with center C and radius r is

$$\|X - C\|^2 = r^2.$$

Point-sphere tests are among the simplest collision tests.

11.3 Conic sections

A general conic in the plane has equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

Depending on the coefficients, this may represent an ellipse, parabola, hyperbola, or degenerate conic.

In matrix form, let

$$x = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad q = \begin{pmatrix} D \\ E \end{pmatrix}.$$

Then the conic equation is

$$x^T Q x + q^T x + F = 0.$$

11.4 Quadrics

A quadric in $\mathbb{R}^3$ is given by a second-degree equation:

$$x^T Q x + q^T x + r = 0,$$

where Q is a symmetric 3×3 matrix. Spheres, ellipsoids, paraboloids, hyperboloids, and cones are examples.

Computer science viewpoint

Implicit surfaces such as spheres and quadrics are used in ray tracing, collision detection, constructive solid geometry, and geometric modeling. The equation $F(x) = 0$ describes the surface; the sign of $F(x)$ often describes inside versus outside.

12 Convexity and half-space geometry

12.1 Convex sets

Definition 12.1 (Convex set)**.** A set $S \subseteq \mathbb{R}^n$ is convex if for all $x, y \in S$ and all $t \in [0, 1]$,

$$(1 - t)x + ty \in S.$$

In words, the line segment between any two points of the set lies entirely inside the set. Examples of convex sets include:

- lines and planes,
- segments,
- disks and balls,
- half-spaces,
- intersections of convex sets,
- convex polygons and convex polyhedra.

12.2 Convex hull

Definition 12.2 (Convex hull)**.** The convex hull of a set $S \subseteq \mathbb{R}^n$, denoted $\operatorname{conv}(S)$, is the smallest convex set containing S.

For finitely many points $p_1, \ldots, p_m$, the convex hull is the set of all convex combinations:

$$\operatorname{conv}\{p_1, \ldots, p_m\} = \left\{ \sum_{i=1}^{m} \alpha_i p_i : \alpha_i \geq 0, \sum_{i=1}^{m} \alpha_i = 1 \right\}.$$

12.3 Convex polygons as intersections of half-planes

A convex polygon can be described either by its vertices or by inequalities:

$$P = \{x \in \mathbb{R}^2 : Ax \leq b\}.$$

This representation is useful in clipping, linear programming, visibility, and collision detection.

12.4 Separating hyperplanes

A central idea in convex geometry is that disjoint convex sets can often be separated by a hyperplane. Informally, if two closed convex sets do not overlap, there may exist a vector n and scalar c such that

$$n \cdot x + c \leq 0$$

for all points in one set and

$$n \cdot y + c \geq 0$$

for all points in the other set.

This idea appears in optimization, support vector machines, collision detection, and computational geometry.

13 Geometry of algorithms

13.1 Nearest neighbor and distance geometry

Many algorithms depend on distances. Given points $p_1, \ldots, p_m$ and a query point q, the nearest neighbor problem asks for

$$\operatorname*{argmin}_i \|q - p_i\| .$$

Since the square root is monotone, it is often better computationally to minimize squared distance:

$$\operatorname*{argmin}_i \|q - p_i\|^2 .$$

This avoids computing square roots.

Computer science viewpoint

In performance-critical code, compare squared distances rather than distances when only ordering is needed.

13.2 Voronoi cells

Given sites $p_1, \ldots, p_m$, the Voronoi cell of p_i is

$$V_i = \{x : \|x - p_i\| \leq \|x - p_j\| \text{ for all } j\}.$$

For two sites p_i, p_j, the boundary between their regions is defined by

$$\|x - p_i\|^2 = \|x - p_j\|^2 .$$

Expanding:

$$(x - p_i) \cdot (x - p_i) = (x - p_j) \cdot (x - p_j).$$

So

$$x \cdot x - 2p_i \cdot x + p_i \cdot p_i = x \cdot x - 2p_j \cdot x + p_j \cdot p_j.$$

Cancel $x \cdot x$:

$$2(p_j - p_i) \cdot x = p_j \cdot p_j - p_i \cdot p_i.$$

This is a hyperplane. Thus Voronoi cells in Euclidean space are intersections of half-spaces.

13.3 Ray casting

A ray in $\mathbb{R}^3$ is

$$r(t) = O + td, \qquad t \geq 0.$$

Ray casting and ray tracing repeatedly solve intersection problems:

- ray-plane intersection,
- ray-triangle intersection,
- ray-sphere intersection,
- ray-box intersection.

For a sphere $\|X - C\|^2 = r^2$, substitute $X = O + td$:

$$\|O + td - C\|^2 = r^2.$$

Let $m = O - C$. Then

$$\|m + td\|^2 = r^2.$$

Expanding:

$$(d \cdot d)t^2 + 2(m \cdot d)t + (m \cdot m - r^2) = 0.$$

This is a quadratic equation in t. Real nonnegative roots correspond to ray-sphere intersections.

13.4 Normals and lighting

For a surface $F(x, y, z) = 0$, the gradient

$$\nabla F = (F_x, F_y, F_z)$$

is normal to the surface, assuming it is nonzero. For a sphere

$$F(X) = \|X - C\|^2 - r^2,$$

we get

$$\nabla F(X) = 2(X - C).$$

Therefore the outward unit normal at X is

$$N = \frac{X - C}{\|X - C\|}.$$

Computer science viewpoint

Normals are essential in shading, collision response, surface reconstruction, and physics engines. The dot product between a normal and a light direction determines basic diffuse lighting.

14 Numerical issues in analytic geometry

14.1 Floating-point arithmetic

In exact mathematics, we can test whether

$$\det(B - A, C - A) = 0$$

to decide collinearity. In floating-point computation, the result may be very close to zero without being exactly zero.

A common practical test is

$$|\det(B - A, C - A)| < \varepsilon,$$

where $\varepsilon > 0$ is a tolerance. However, choosing ε is not trivial: too large gives false collinearity, too small misses near-degenerate cases.

14.2 Scaling matters

A fixed tolerance can fail when coordinates are very large or very small. For example, an error of 10^{-9} may be huge for coordinates around 10^{-12}, but negligible for coordinates around 10^{6}. Relative tolerances are often safer.

14.3 Degenerate cases

Robust geometry code must handle cases such as:

- zero-length segments,
- parallel lines,
- coincident lines,
- points exactly on boundaries,
- nearly collinear triples,
- singular transformation matrices,
- triangles with nearly zero area.

Common pitfall

Most geometric algorithms are simple in the generic case and difficult in boundary cases. Always decide what should happen on boundaries before implementing the algorithm.

14.4 Stable formulas

A formula may be mathematically correct but numerically unstable. Good computational geometry prefers formulas that avoid unnecessary subtraction of nearly equal numbers, avoid division by small quantities, and handle degeneracy explicitly.

For example, to compare distances, prefer

$$\|x - y\|^2 < \|x - z\|^2$$

instead of computing both square roots.

15 Worked examples

15.1 Example 1: distance from point to line

Let

$$P = (1, 2), \qquad Q = (4, 3), \qquad R = (3, 6).$$

Find the distance from R to the line through P and Q.

The direction vector is

$$d = Q - P = (3, 1).$$

The line is

$$L = P + td.$$

The projection parameter is

$$t^* = \frac{d \cdot (R - P)}{d \cdot d}.$$

Compute

$$R - P = (2, 4),$$
$$d \cdot (R - P) = 3 \cdot 2 + 1 \cdot 4 = 10,$$
$$d \cdot d = 3^2 + 1^2 = 10.$$

Thus

$$t^* = 1.$$

The closest point is

$$C = P + d = (4, 3) = Q.$$

Therefore

$$\operatorname{dist}(R, L) = \|R - C\| = \|(3, 6) - (4, 3)\| = \sqrt{1 + 9} = \sqrt{10}.$$

15.2 Example 2: equation of a plane

Find the plane through

$$A = (1, 0, 0), \qquad B = (0, 1, 0), \qquad C = (0, 0, 1).$$

We have

$$u = B - A = (-1, 1, 0), \qquad v = C - A = (-1, 0, 1).$$

A normal vector is

$$n = u \times v.$$

Compute

$$u \times v = \begin{pmatrix} 1 \cdot 1 - 0 \cdot 0 \\ 0 \cdot (-1) - (-1) \cdot 1 \\ (-1) \cdot 0 - 1 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The plane is

$$n \cdot (X - A) = 0.$$

Thus

$$(1, 1, 1) \cdot (x - 1, y, z) = 0,$$

so

$$x + y + z - 1 = 0.$$

15.3 Example 3: barycentric point test

Let

$$A = (0, 0), \quad B = (4, 0), \quad C = (0, 3), \quad P = (1, 1).$$

Write

$$P = A + \beta(B - A) + \gamma(C - A).$$

Since $A = (0, 0)$,

$$(1, 1) = \beta(4, 0) + \gamma(0, 3) = (4\beta, 3\gamma).$$

Thus

$$\beta = \frac{1}{4}, \qquad \gamma = \frac{1}{3}.$$

Then

$$\alpha = 1 - \frac{1}{4} - \frac{1}{3} = \frac{5}{12}.$$

All barycentric coordinates are positive, so P lies inside the triangle.

15.4 Example 4: affine transformation of a point

Let

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

Then

$$T(x) = Ax + b = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 9 \\ 11 \end{pmatrix}.$$

In homogeneous coordinates,

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 3 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ 11 \\ 1 \end{pmatrix}.$$

16 Summary of core formulas

Vectors and distances

$$\|v\| = \sqrt{v_1^2 + \cdots + v_n^2}, \qquad \operatorname{dist}(P, Q) = \|Q - P\|.$$

$$u \cdot v = \sum_i u_i v_i, \qquad \cos\theta = \frac{u \cdot v}{\|u\| \, \|v\|}.$$

$$\operatorname{proj}_a b = \frac{a \cdot b}{a \cdot a} a.$$

Lines, planes, and hyperplanes

Line:

$$X = P + td.$$

Plane in $\mathbb{R}^3$:

$$X = P + su + tv, \qquad n \cdot (X - P) = 0.$$

Hyperplane:

$$n \cdot x + c = 0.$$

Projection onto hyperplane:

$$p = q - \frac{n \cdot q + c}{\|n\|^2} n.$$

Determinants and orientation

$$\det(u, v) = u_x v_y - u_y v_x.$$

$$\text{orient}(A, B, C) = \det(B - A, C - A).$$

Polygon area:

$$A = \frac{1}{2} \left| \sum_{i=0}^{m-1} \det(p_i, p_{i+1}) \right|.$$

Affine transformations

$$T(x) = Ax + b.$$

Homogeneous form in 2D:

$$\begin{pmatrix} A & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} Ax + b \\ 1 \end{pmatrix}.$$

17 Exercises

Basic exercises

1. Let $u = (3, -1, 2)$ and $v = (1, 4, -2)$. Compute $u + v$, $2u - v$, $u \cdot v$, and $\|u\|$.
2. Find the distance between $P = (2, -1)$ and $Q = (-3, 5)$.
3. Determine whether $u = (2, 1)$ and $v = (-3, 6)$ are perpendicular.
4. Compute the projection of $b = (4, 3)$ onto $a = (1, 2)$.
5. Find a parametric equation and an implicit equation of the line through $(1, 2)$ and $(4, -1)$.
6. Determine whether the points $(0, 0)$, $(2, 1)$, and $(4, 2)$ are collinear using a determinant.
7. Compute the signed area of the triangle with vertices $(0, 0)$, $(3, 0)$, and $(0, 2)$.
8. Find the equation of the plane through $(1, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 3)$.

Intermediate exercises

1. Derive the formula for distance from a point to a hyperplane in $\mathbb{R}^n$.
2. Given a segment $[A, B]$ and point Q, prove why clamping the projection parameter gives the closest point on the segment.
3. Let $A = (0, 0)$, $B = (5, 0)$, $C = (1, 3)$. Compute barycentric coordinates of $P = (2, 1)$. Is P inside the triangle?
4. Find the intersection point of lines $P + td$ and $Q + su$, where $P = (0, 0)$, $d = (2, 1)$, $Q = (1, 3)$, and $u = (1, -1)$.
5. Show that an affine map preserves midpoints: if $M = (P + Q)/2$, then $T(M) = (T(P) + T(Q))/2$.
6. Write a homogeneous matrix that rotates by $90°$, then translates by $(3, -2)$.
7. Let $r(t) = O + td$ be a ray and let $\|X - C\|^2 = R^2$ be a sphere. Derive the quadratic equation for ray-sphere intersection.

Computer science oriented exercises

1. Implement the orientation test. Then use it to determine whether two non-collinear segments intersect.
2. Implement point-to-segment distance and test all boundary cases: point closest to the interior, closest to the first endpoint, and closest to the second endpoint.
3. Given triangle vertices and colors at vertices, use barycentric coordinates to interpolate the color at an interior point.
4. Derive the hyperplane that separates two points p and q into equal-distance regions. Interpret the result as a Voronoi boundary.
5. Given a camera frame with origin O and orthonormal axes r, u, f, write the matrix that converts world coordinates to camera coordinates.
6. Consider a linear classifier $w \cdot x + b = 0$. Show that the signed distance from a point x to the decision boundary is $(w \cdot x + b)/\|w\|$.
7. Explain why comparing squared distances is usually better than comparing distances in nearest-neighbor search.

18 Short answers to selected exercises

1. For $u = (3, -1, 2)$, $v = (1, 4, -2)$:
$$u + v = (4, 3, 0), \quad 2u - v = (5, -6, 6), \quad u \cdot v = 3 - 4 - 4 = -5, \quad \|u\| = \sqrt{14}.$$
2. For $P = (2, -1)$, $Q = (-3, 5)$:
$$\operatorname{dist}(P, Q) = \sqrt{(-5)^2 + 6^2} = \sqrt{61}.$$
3. $(2, 1) \cdot (-3, 6) = -6 + 6 = 0$, so the vectors are perpendicular.
4. For $a = (1, 2)$, $b = (4, 3)$:
$$\operatorname{proj}_a b = \frac{a \cdot b}{a \cdot a} a = \frac{10}{5}(1, 2) = (2, 4).$$
5. Through $(1, 2)$ and $(4, -1)$, direction is $(3, -3)$, so
$$X = (1, 2) + t(3, -3).$$
A normal is $(1, 1)$, so
$$(1, 1) \cdot ((x, y) - (1, 2)) = 0,$$
hence
$$x + y - 3 = 0.$$

Closing remark

Analytic geometry is one of the most useful mathematical toolkits for computer science because it converts geometry into computation. A line becomes a parameterized set. A plane becomes a normal vector and an equation. A distance becomes a norm. An angle becomes a dot product. An area becomes a determinant. A transformation becomes a matrix. These translations are the foundation of many practical systems: rendering engines, simulators, geographic software, robotics, computer vision, CAD systems, machine learning models, and numerical algorithms.

The essential habit is to move fluently between pictures, formulas, and algorithms. A geometric idea is fully understood when one can draw it, express it algebraically, prove the formula, and implement it robustly.